Vanishing Gradient

- Gating mechanism
- Attention

↑

Sequence 2 Sequence problem

LSTM

Long Short-term Memory

↑

Gate || Cell State

hidden State

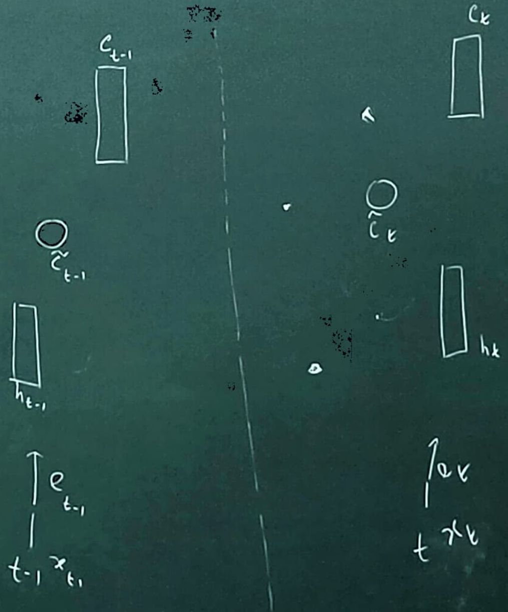
Cell Content $\tilde{C}$

0 1

Forget

Input

Output



Vanishing Gradient

- Gating mechanism
- Attention

↑

Sequence 2 Sequence problem

LSTM

Long Short-term Memory

Gate || Cell State

hidden State

Cell Content $\tilde{c}$

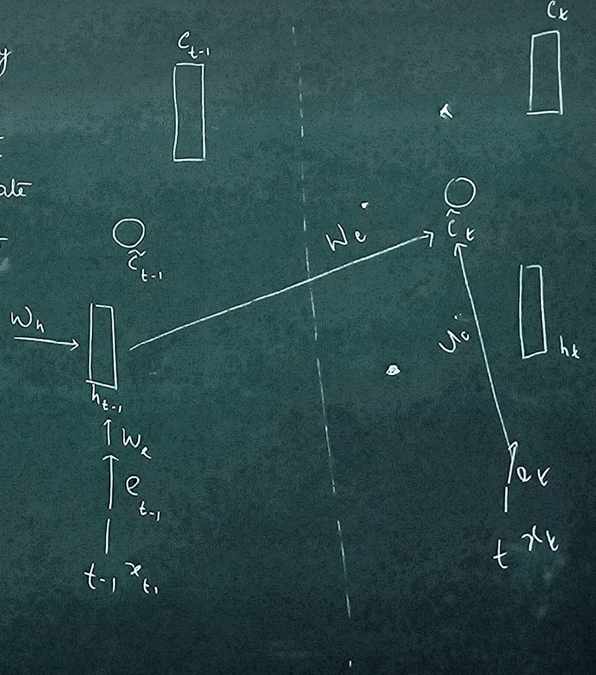
0 1

0 0

Forget (f)

Input (i)

Output (o)



Vanishing Gradient

- Gating mechanism
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↑

Sequence 2 Sequence problem

LSTM

Long Short-term **Memory**

↑

Gate || Cell State

hidden State

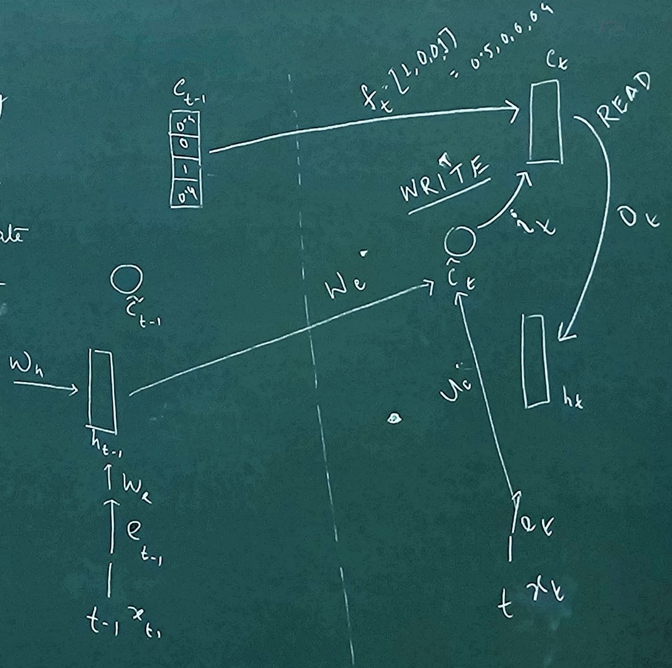
0 1

Cell Content $\tilde{c}$

Forget (f)

Input (i)

Output (o)



Vanishing Gradient

- Gating mechanism
- Attention

↑

Sequence 2 Sequence problem

LSTM

Long Short-term Memory

↑

Gate || Cell State

hidden State

0 1

0 0

Cell Content $\tilde{c}$

Forget (f)

Input (i)

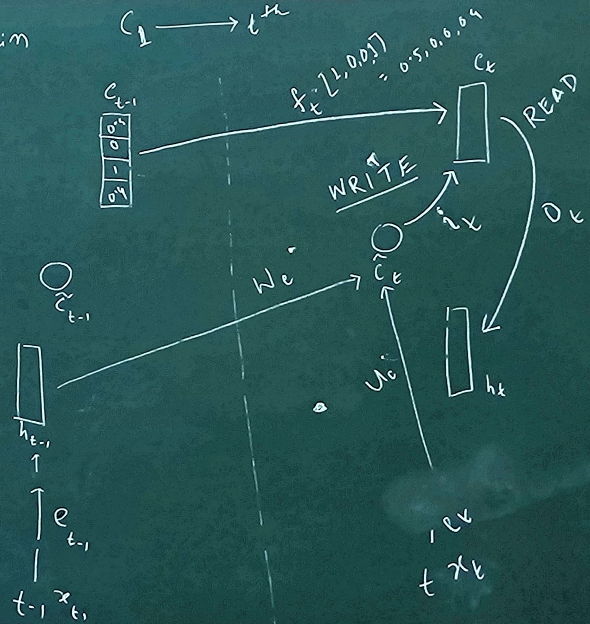
Output (o)

$$f = F_1(h_{t-1}, e_t) = W_f h_{t-1} + U_f e_t + b_f$$

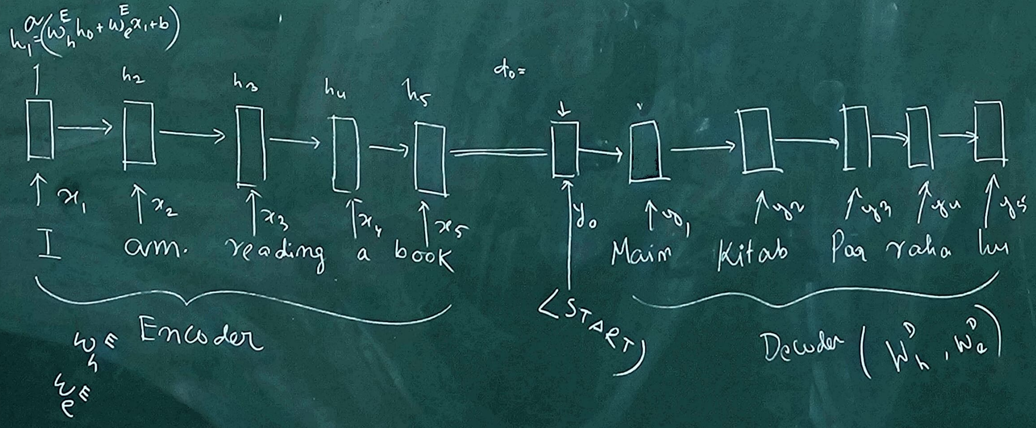
$$i = F_2(h_{t-1}, e_t) = W_i h_{t-1} + U_i e_t + b_i$$

$$o = F_3(h_{t-1}, e_t) = W_o h_{t-1} + U_o e_t + b_o$$

retain

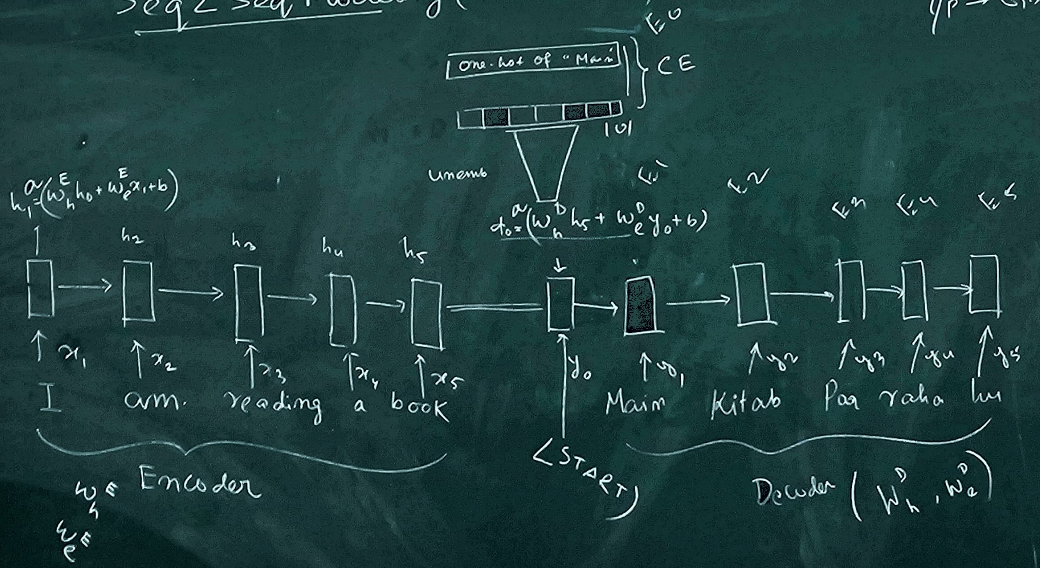


Seq2Seq Modeling (En-De Model)

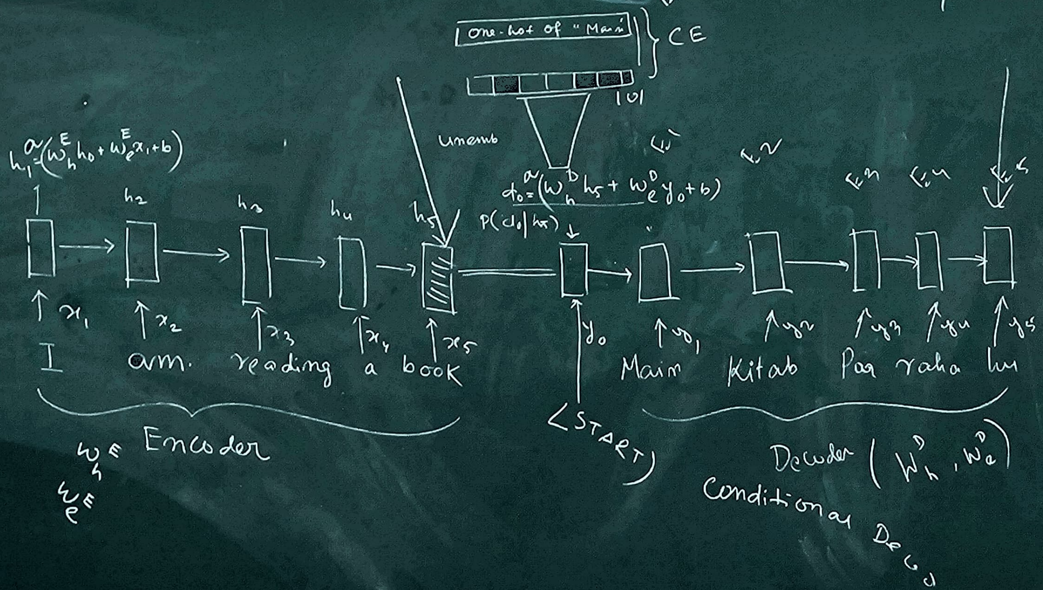


Seq2 Seq Modeling (En-De Model)

I/p → Ground-Truth → Teacher Forcing



Seq2Seq Modeling (En-De Model)



I/p → Ground-Truth → Teacher Forcing

End-to-End Training

1) Vanishing Gradient
2) Seq modeling - slow
3) Bottleneck prob

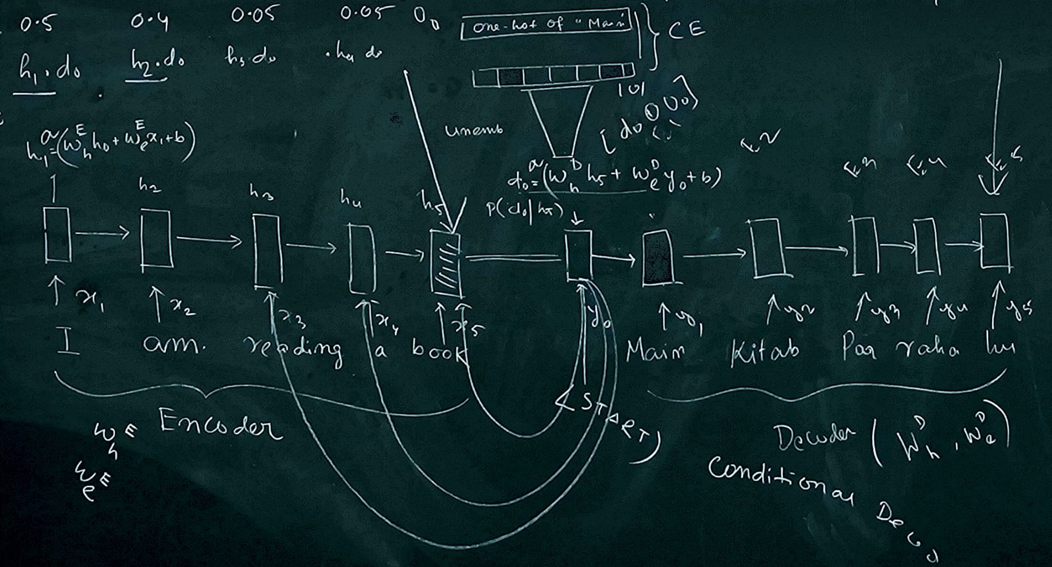
$0.5 \times h_1 + 0.4 h_2 + 0.05 h_3 + 0.05 h_4$

Attn dist	0.5	0.4	0.05	0.05
Attn score	$h_1 \cdot d_0$	$h_2 \cdot d_0$	$h_3 \cdot d_0$	$h_4 \cdot d_0$

$-\infty \leftrightarrow +\infty$

$h_1 = \sigma(W_h^E h_0 + W_e^E x_1 + b)$

$d_0 = \sigma(W_h^D h_5 + W_e^D y_0 + b)$



I/P → Ground-Truth → Teacher Forcing

End-to-End Training

1) Vanishing Gradient
2) Seq modeling – slow
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